

Congratulations on getting this assignment! It's a comprehensive project that evaluates your skills in backend development, API integrations, frontend UI, and cloud deployment. Let's break down the requirements and provide a step-by-step guide to help you complete this assignment.

****Backend Development (Node.js and Express)****

1. Develop a REST API using Node.js and Express:

- * Create a new Node.js project and install Express using npm.
- * Set up a MongoDB database (or any other database of your choice) and connect it to your Node.js project.

Implement CRUD operations for charging stations:

- + Create a new charging station: Use the 'POST' method to create a new charging station.
- + Read all charging stations: Use the 'GET' method to retrieve all charging stations.
- + Update a charging station: Use the 'PUT' method to update an existing charging station.
- + Delete a charging station: Use the 'DELETE' method to delete a charging station.

2. Implement User Authentication using JWT:

- * Create a register API (signup) to create a new user.
- * Create a login API to issue a JWT token.
- * Protect routes for managing charging stations using the JWT token.

****Frontend Development (Vue.js)****

1. Create a login screen that interacts with the backend API:

- * Use Vue.js to create a login screen that sends a request to the backend API for authentication.
 - * Use the JWT token to authenticate and authorize requests to the backend API.
2. Build a Charger Listing Page:
- * Use Vue.js to create a